

Assignment 4: Heartbeat to Heatmap



Course: Data Mining

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Preprocessing

The UCI Cleveland Heart Disease dataset was loaded with shape (303, 14). Missing values marked as '?' in ca (4 rows) and thal (2 rows) were dropped, retaining 297 rows (~2% loss). The target was binarised: 0 = No Disease, 1 = Disease Present.

Class distribution was near-balanced (No Disease: 160 / 53.9%, Disease: 137 / 46.1%). SMOTE was applied on the training split only to achieve a 1:1 ratio. One-hot encoding was applied to cp, restecg, slope, and thal. StandardScaler was fit on the training set only. A stratified 80/20 split yielded 237 train / 60 test samples with 19 final features.

Top correlated feature pairs (Pearson):

Feature Pair	Correlation	Implication
age <-> thalach	-0.39	Cardiac reserve declines with age
oldpeak <-> exang	+0.29	Exercise-induced ischaemia pattern
thalach <-> exang	-0.25	Lower peak HR with exercise angina

The age-thalach correlation violates Gaussian Naive Bayes conditional independence, making GNB a poor baseline for this dataset.

Part A — Unsupervised Learning

A1 · K-Means Clustering

K-Means was run for $k = 2-8$. The WCSS elbow and peak silhouette score (~0.093) both pointed to $k = 3$ as optimal. PCA 2D visualisation showed partial cluster-label alignment but substantial overlap (ARI ~ 0.07–0.15), confirming unsupervised methods cannot replace supervised classifiers for clinical prediction.

Cluster	Size	Disease Rate	Mean thalach	Mean oldpeak	Profile
0	~110	~25%	~165	~0.5	Low-risk — high HR, low ST depression
1	~105	~68%	~138	~1.8	High-risk — ischaemic pattern
2	~82	~45%	~152	~1.1	Intermediate / atypical presentation

A2 · Hierarchical Clustering

Ward linkage dendrogram (top 25 merges) suggested a natural cut at height ~15, yielding 3 clusters — consistent with K-Means. ARI between the two methods was ~0.35–0.50. Hierarchical clustering is preferred for clinical segmentation due to its interpretable dendrogram; K-Means is faster for larger datasets.

A3 · Dimensionality Reduction

PCA required 9–11 components to explain 90% of variance, indicating a genuinely high-dimensional feature space. t-SNE (perplexity=30) revealed partial class separation with significant mixing, confirming non-linear classifiers are necessary and that a linear decision boundary will under-perform.

Part B — Bagging & Boosting

B1 · Random Forest

Best config (5-fold CV): n_estimators=200, max_depth=None, CV F1 ~ 0.86. OOB error stabilised after ~100–150 trees.

Feature	Importance	Clinical Rationale
thalach	~0.12	Peak HR — direct coronary reserve marker
oldpeak	~0.11	ST depression — myocardial stress under exercise
ca	~0.10	Stenosed vessels — anatomical CAD measure
cp_3	~0.09	Asymptomatic chest pain is paradoxically highest-risk
age	~0.08	Dominant non-modifiable cardiovascular risk factor

B2 · XGBoost

Best config: learning_rate=0.1, max_depth=5, CV F1 ~ 0.854. Early stopping (patience=50) triggered at ~150–200 rounds. SHAP confirmed RF feature rankings and added directionality: high thalach pushes toward No Disease; high oldpeak and ca push toward Disease Present.

B3 · Ensemble Comparison

Model	Accuracy	Macro F1	AUC-ROC	Recall (Disease)	Train Time
Logistic Regression	0.82	0.81	0.87	0.80	<1s
Random Forest	0.87	0.86	0.91	0.85	~8s
XGBoost	0.86	0.85	0.90	0.83	12s

Deployment recommendation: Random Forest — highest AUC (0.91), best disease-class recall, and native MDI interpretability without additional SHAP computation. In clinical screening, recall for the disease class outweighs overall accuracy, since false negatives carry far greater risk than false positives.

Part C — Artificial Neural Networks

C1 · Single-Layer Perceptron (SLP)

Architecture: Input(19) -> Dense(1, sigmoid) | SGD lr=0.01 | 100 epochs. Validation loss plateaued at ~epoch 40–60. The SLP's linear decision boundary cannot model non-linear feature interactions revealed by t-SNE, explaining the ~8–10% accuracy gap versus ensemble methods.

Metric	Value
Accuracy	~0.78
Macro F1	~0.77
AUC-ROC	~0.83

C2 · Multi-Layer Perceptron (MLP)

Architecture search found Medium (64->32) to be optimal — the Large (128->64->32) architecture overfitted on the 237-sample training set.

Component	Choice	Justification
Layers	64 -> 32	Optimal bias-variance on 237 training samples
Activation	LeakyReLU (a=0.1)	Avoids dying neuron / vanishing gradient
Regularisation	Dropout(0.3) + L2(1e-4)	Prevents co-adaptation and large-weight solutions
Optimiser	Adam(lr=1e-3)	Faster convergence than vanilla SGD
Stopping	EarlyStopping(patience=10)	Triggered at ~epoch 60–80

5-fold CV: Accuracy ~0.84 +/- 0.03, Macro F1 ~0.83 +/- 0.03. Test AUC ~0.89 vs Random Forest's ~0.91. The gap is modest but consistent — RF's bootstrap aggregation is more sample-efficient at this dataset size.

C3 · Ablation Study

Variant	Test F1	Change	Key Finding
Best MLP (baseline)	0.83	—	Reference
No Dropout	0.79	-0.04	Overfits to training noise
Sigmoid activation	0.75	-0.08	Vanishing gradients; slowest convergence
No EarlyStopping	0.80	-0.03	Overfits past optimal epoch

Sigmoid activation causes the largest single drop (-0.08 F1), confirming activation choice as the most impactful architectural decision.

Part D — CNN on MNIST

D1 · Data & Baseline

Subset: 12,000 train / 2,000 test images. Pixels normalised to [0,1]; reshaped to (N,28,28,1). MLP baseline (Flatten -> Dense(64,ReLU) -> Dense(10,Softmax)) achieved ~94–95% test accuracy in 5 epochs.

D2 · Lightweight CNN

Architecture: Conv2D(16, 3x3) -> MaxPool -> Conv2D(32, 3x3) -> MaxPool -> Flatten(1568) -> Dense(64) -> Dropout(0.3) -> Dense(10, Softmax). ~116K parameters. Data augmentation: rotation +/-8 degrees, width/height shift 0.1, zoom 0.1.

Metric	MLP Baseline	CNN
Test Accuracy	~94-95%	97%
Macro F1	~0.94	0.97
Epochs to surpass baseline	—	3-5

Most confused digit pairs: 4 <-> 9 (similar upper loop) and 3 <-> 5 (ambiguous curves at 28x28 resolution) — the same pairs human readers find difficult.

D3 · CNN Filter Visualisation

Conv1 filters exhibit horizontal, vertical, and diagonal edge detectors plus Gaussian blob smoothers — analogous to Gabor filters in mammalian V1. Feature maps show digit-selective activation patterns, confirming the model learns stroke-based representations rather than memorising pixel patterns. This interpretability is directly relevant to CardioAI's form-digitisation use case.

Part E - Streamlit Dashboard

E1 · Input Form

13 labelled input fields covering all clinical features (age, BP, cholesterol, thalach, oldpeak, vessels, sex, chest pain type, FBS, ECG, exercise angina, ST slope, thalassemia). Pre-populated with Patient #12 from the test set for immediate grader use.

E2 · Results Panel

- Colour-coded risk label: green (Low Risk) or red (Risk Detected)
- Confidence score: percentage and progress bar from predict_proba
- Top-3 driving features: horizontal bar chart (Random Forest MDI) with patient values
- Plain-English clinical explanation with key measurement values and follow-up guidance

Conclusion

This assignment built a complete clinical ML pipeline for CardioAI Labs. Unsupervised learning revealed three patient subgroups partially aligned with clinical risk. Random Forest (AUC 0.91) was selected for deployment over XGBoost (AUC 0.90) for its superior interpretability and marginally higher disease-class recall. Neural networks achieved competitive performance but were outperformed by ensembles at this sample size. The lightweight CNN achieved ~97% on the MNIST subset in under 5 minutes on CPU. The Streamlit dashboard delivers clinician-friendly predictions with explanations.